

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Two identical rectangular prisms each have a height of **90 centimeters (cm)**. The base of each prism is a square, and the surface area of each prism is $K \text{ cm}^2$. If the prisms are glued together along a square base, the resulting prism has a surface area of $\frac{92}{47} K \text{ cm}^2$. What is the side length, in **cm**, of each square base?

Answer

- A. 4
- B. 8
- C. 9
- D. 16

Correct Answer: B

Rationale

Choice B is correct. Let x represent the side length, in **cm**, of each square base. If the two prisms are glued together along a square base, the resulting prism has a surface area equal to twice the surface area of one of the prisms, minus the area of the two square bases that are being glued together, which yields $2K - 2x^2 \text{ cm}^2$. It's given that this resulting surface area is equal to $\frac{92}{47} K \text{ cm}^2$, so $2K - 2x^2 = \frac{92}{47} K$. Subtracting $\frac{92}{47} K$ from both sides of this equation yields $2K - \frac{92}{47} K - 2x^2 = 0$. This equation can be rewritten by multiplying $2K$ on the left-hand side by $\frac{47}{47}$, which yields $\frac{94}{47} K - \frac{92}{47} K - 2x^2 = 0$, or $\frac{2}{47} K - 2x^2 = 0$. Adding $2x^2$ to both sides of this equation yields $\frac{2}{47} K = 2x^2$. Multiplying both sides of this equation by $\frac{47}{2}$ yields $K = 47x^2$. The surface area K , in cm^2 , of each rectangular prism is equivalent to the sum of the areas of the two square bases and the areas of the four lateral faces. Since the height of each rectangular prism is **90 cm** and the side length of each square base is $x \text{ cm}$, it follows that the area of each square base is $x^2 \text{ cm}^2$ and the area of each lateral face is $90x \text{ cm}^2$. Therefore, the surface area of each rectangular prism can be represented by the expression $2x^2 + 4(90x)$, or $2x^2 + 360x$. Substituting this expression for K in the equation $K = 47x^2$ yields $2x^2 + 360x = 47x^2$. Subtracting $2x^2$ and $360x$ from both sides of this equation yields $0 = 45x^2 - 360x$. Factoring x from the right-hand side of this equation yields $0 = x(45x - 360)$. Applying the zero product property, it follows that $x = 0$ and $45x - 360 = 0$. Adding 360 to both sides of the equation $45x - 360 = 0$ yields $45x = 360$. Dividing both sides of this equation by 45 yields $x = 8$. Since a side length of a rectangular prism can't be **0**, the length of each square base is **8 cm**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.